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Chapter 7

Designing Effective Feedback Messages in Serious Games and Simulations: A Research Review

Cheryl I. Johnson, Shannon K.T. Bailey, and Wendi L. Van Buskirk

Abstract Taking a value-added approach, we examined the impact of feedback on learning outcomes and performance in serious games and simulations. Although feedback has been demonstrated to be beneficial in traditional learning environments, we explore how feedback has been implemented in game- and simulation-based learning environments. In this review, we discuss critical characteristics that affect the efficacy of feedback, including the content of feedback messages, the modality in which feedback is presented, the timing of feedback presentation, and learner characteristics. General guidelines based on the research evidence are provided, and the theoretical implications are discussed in the context of the cognitive theory of multimedia learning (CTML; The Cambridge handbook of multimedia learning, Mayer, 2014b).

Keywords Feedback • Modality of feedback • Timing of feedback • Content of feedback • Adaptation

7.1 Introduction

Over the last decade, the promise of games and simulations for education and training has been touted by researchers and practitioners alike. Many boast about the motivational potential of games compared to traditional classroom instruction and argue that playing games is more intrinsically motivating. It follows that players will devote more cognitive effort while they are playing and are more likely to play

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again, leading to higher learning outcomes (Garris, Ahlers, & Driskell, 2002; Gee, 2007; Prensky, 2001). There have been several meta-analyses examining whether or not games and simulations truly are more effective for learning than conventional methods of instruction (e.g., Hays, 2005; Ke, 2009; O'Neil & Perez, 2008; Sitzmann, 2011; Vogel et al., 2006; Wouters, van Nimwegen, van Oostendorp, & van der Speck, 2013), and the findings have been mixed. However, several reviews suggest that games and simulations are more effective for learning when they include instructional support (e.g., Hays, 2005; Ke, 2009; Sitzmann, 2011; Wouters et al., 2013; Wouters & van Oostendorp, 2013). The focus of this chapter is to explore what we know about feedback, arguably one of the most important instructional features in serious games.

7.1.1 *Defining Serious Games*

A *serious game* is a game that is played with the intention to learn knowledge or a skill. Serious games are referred to by many names, including educational games, simulation games, game-based learning, and games for learning. The serious game literature is riddled with inconsistent definitions. There is no clear consensus in the extant literature on the definition of a game. Several definitions do converge on some of the defining characteristics of games that include: games are interactive (the player can make actions), rule based (events are bound within a set of rules), challenging (opportunities exist for success and for overcoming difficulties), responsive (the game responds to player actions), and goal oriented (the player works to achieve a knowable goal and can monitor progress along the way; Garris et al., 2002; Mayer, 2014a; Mayer & Johnson, 2010; Sitzmann, 2011; Vogel et al., 2006; Wouters et al., 2013).

Further complicating the issue is that the line between games and simulations is not always well-defined (Sitzmann, 2011; Vogel et al., 2006). *Simulations* can be defined as representations of a real-world system or process that allow the learner to test variables to see how the system will respond or to practice certain behaviors or skills that may be dangerous or cost-prohibitive to perform in real life (Merchant, Goetz, CiFuentes, Keeney-Kennicutt, & Davis, 2014; Sitzmann, 2011; Vogel et al., 2006; Wouters et al., 2013). This definition of a simulation shares several common attributes with that of serious games, including interactivity, challenge, and responsiveness. Following a similar logic, previous researchers have decided to combine serious games and simulations within their analyses (Garris et al., 2002; Sitzmann, 2011; Wouters et al., 2013), and we have chosen to do so here as well. Since it is not the goal of this chapter to resolve this debate, for simplicity's sake, we include game and simulation studies in our analysis and draw those distinctions when possible.

Not surprisingly, the serious game and simulation literature is vast and diverse. To address this diversity in the literature, Mayer and Johnson (2010) identified three different approaches to empirical games research that vary in terms of research goals. The goal of the *media comparisons* approach is to examine whether people

learn better by playing games or learning from conventional instruction. The goal of the *cognitive consequences* approach is to investigate if playing a particular game causes players to improve one or more cognitive skills, such as spatial ability, perceptual attention, or motor skills. The goal of the *value-added* approach is to examine if adding an instructional feature to the game improves learning compared to a group who plays the same game but without the added instructional feature. This chapter will focus on research that takes a value-added approach to determine situations in which feedback improves student learning.

7.1.2 *Defining Feedback*

The purpose of feedback in serious games and simulations is to direct learners to improve performance, motivation, or learning outcomes by various methods of providing information to learners about the correctness of their responses (Shute, 2008). Providing feedback allows a learner to evaluate his or her progress and responses, identify knowledge gaps, and repair faulty knowledge (Johnson & Priest, 2014). From a broad review of the feedback literature focusing on serious games, simulations, and multimedia learning, we have identified four characteristics of feedback under which most feedback types fall. Methods of providing feedback vary by (1) type of feedback given (e.g., procedural or conceptual information, amount of elaboration), (2) the timing of feedback after a response (e.g., immediately following a question, at the end of a practice session), (3) the modality in which information is presented (e.g., spoken feedback, text-based feedback), and by (4) adapting to learner characteristics (e.g., prior knowledge, spatial ability). Feedback has been demonstrated to be an effective instructional technique to benefit learning (Azevedo & Bernard, 1995), and several meta-analyses have been conducted to determine when, how, and why it works (Bangert-Drowns, Kulik, Kulik, & Morgan, 1991; Hattie & Timperley, 2007; Kluger & DeNisi, 1996; Narciss & Huth, 2004; Shute, 2008). Although feedback has been studied mostly in more traditional educational settings, recent meta-analyses taking a value-added approach have found that adding instructional support, such as feedback, to a serious game improves learning outcomes (Ke, 2009; Mayer, 2014a; Wouters & van Oostendorp, 2013).

7.1.3 *Present Chapter*

The goal of this chapter is to address what is currently known about feedback in the serious games and simulations literature and draw conclusions about feedback strategies that can be useful for improving learning in the design and research of serious games. Throughout the chapter, we present empirical evidence from studies that take a value-added perspective to illustrate when certain feedback messages may be more effective than others. Although we originally set out to investigate the impact

of feedback in serious games on learning *and* motivation, we found little empirical evidence on how providing feedback affects motivation. Therefore, since the literature on feedback and motivation is not yet mature enough to draw any strong conclusions, we chose to focus on the effects of different feedback strategies on performance and learning outcomes.

In this chapter, we outline the theoretical perspectives motivating the incorporation of feedback into games, highlight the core characteristics of feedback (content, modality, timing, and adaptation to individual differences), and provide practical implications of this research for future research and games design.

7.2 Theoretical Motivation

In order to understand how individuals learn from games—and how feedback can be effective in games—we must first consider how learning works. The cognitive theory of multimedia learning (CTML), as depicted in Fig. 7.1, provides a theoretical foundation for understanding how people learn (Mayer, 2014a, 2014b). The CTML has three main assumptions about a learner’s cognitive system. First, learners possess separate channels for processing information: a visual channel and a verbal channel. The visual channel is used to process visual information; in the case of a serious game, learners would utilize the visual channel to view their avatar as they navigate the game environment. Likewise, the verbal channel is used to process verbal information, such as the spoken instructions from a pedagogical agent or teammate. The second assumption is that learners are limited in the amount of cognitive processing that occurs in any one channel at a time. That is, one or both channels can become overloaded if the learner’s cognitive processing demands are too high. The third assumption is that learners are active participants during a learning episode and engage in the cognitive processes of *selecting* the relevant information from the serious game, *organizing* that information into a coherent mental representation, and *integrating* and updating this representation with new incoming information and with prior knowledge stored in long-term memory.

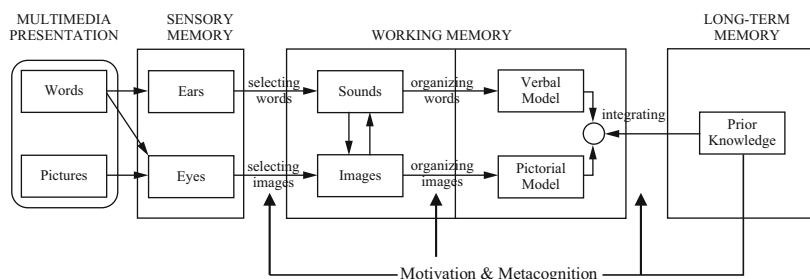


Fig. 7.1 The cognitive theory of multimedia learning (CTML). Adapted from Mayer (2014b)

Additionally, the CTML considers three demands on a learner's cognitive system during a learning episode. *Extraneous processing* is cognitive processing that occurs as a result of poor instructional or game design that does not contribute to the overall educational goal of the serious game or simulation. For example, a game that includes distracting sounds, unnecessary cut scenes, or complicated game controls and interfaces may create a situation of extraneous processing, since these features of the game are not productive toward the learning objectives of the game. *Essential processing* is cognitive processing that is required to mentally represent the information that is being learned in a learner's working memory that is the result of the material's complexity. Consider a learner playing a serious game that teaches how to troubleshoot electric circuits. The learner would need to represent mentally the elements of the circuit and their configuration in order to solve a troubleshooting question. For a novice, this situation is more complex and would likely require more essential processing than if the learner were playing a less complex game, such as a game to learn Spanish-English vocabulary words, which would require holding only two elements in working memory simultaneously. *Generative processing* is the cognitive processing directed at making sense of the essential information being presented to the learner in the serious game including reorganizing it and relating it to prior knowledge, which is dependent on the learner's level of motivation to play the game. If the learner is highly engaged in the serious game, the learner may put more effort into playing than if the game is not very engaging. It is important to reiterate that both essential and generative processing are productive toward the educational goal, while extraneous processing is not. Given that the learner's cognitive system has a limited capacity, extraneous, essential, and generative processing have an additive effect, such that if the extraneous processing demands overwhelm the learner's capacity, the learner will not have the cognitive resources available to engage in the more productive essential and generative processing. This situation can result in lower learning outcomes. Therefore, the goal for serious game designers is to reduce extraneous processing, manage essential processing, and foster generative processing in serious games to promote meaningful learning.

Mayer (2014b) recently expanded the CTML to include motivation and metacognition within the model (see also Moreno & Mayer, 2007). That is, while it is important to consider the types of cognitive processing that occur during learning, it is also important to acknowledge that one must have a desire to engage in the necessary cognitive processing to learn (i.e., motivation) and an understanding of how to manage one's cognitive processing appropriately (i.e., metacognition). Although motivation is also included in the definition of generative processing, the overall role motivation and metacognition play within the CTML model is "underdeveloped" (Mayer, 2014b, p. 66). In Fig. 7.1, an arrow points from long-term memory to the cognitive processes of selecting, organizing, and integrating, illustrating how motivation may affect these processes, and therefore how people learn. More research is needed to define the mechanism by which motivation may affect these processes and how feedback (and other instructional strategies) affects motivation.

Using CTML as a framework, one can consider how feedback could be useful for improving learning under certain conditions and for particular learners. At a high level, feedback serves to aid learners in the process of knowledge construction during a learning episode (Moreno & Mayer, 2005, 2007). In a *discovery-learning* environment, such as a serious game or simulation, the learner may freely explore the environment and receive minimal guidance on what to do and what to learn; in fact, in this free-play type of environment, it is possible that the learner may never come into contact with the to-be-learned material. Such a situation is likely to impose a high extraneous load on the learner, particularly on a novice learner, because he lacks knowledge of what is important to pay attention to within the game (Kirschner, Sweller, & Clark, 2006). Using a *guided* discovery-learning method, such as providing feedback in a serious game, learners can be pointed in the right direction and be provided an assessment of their performance. In addition, with feedback, a learner can also be provided with information about why a certain action or decision was correct or incorrect, and concepts, principles, and strategies can be explicitly explained, so that the learner is not left to make such a discovery on his own. That is, feedback can serve to reduce extraneous processing by guiding the learner to select the appropriate information in the learning environment (Kirschner et al., 2006; Mayer, 2003; Moreno, 2004) and may even serve as a way to help learners with organizing and integrating information (aiding in high level knowledge construction).

7.3 Related Work

7.3.1 Content of Feedback

Feedback content refers to the type and amount of information provided after a learner's response and varies considerably from merely providing a performance score (a form of outcome-based feedback) to providing an explanation of the correct answer (a form of process feedback). Similar to the issues found in the serious games literature, the feedback literature is also plagued with an inconsistent use of terminology. Therefore, in an effort to refine the numerous definitions of feedback content into usable guidelines for serious game designers and researchers, we consolidated and refined terms into a table of feedback content types, definitions, and examples (see Table 7.1). For each type of feedback, we have included alternative names used to refer to similar constructs. The types of feedback are grouped into outcome-based feedback and process-based feedback. *Outcome feedback* informs learners about their performance outcome or learning progress. *Process feedback* directs learners on the processes and/or strategies used to reach the correct answer or action in the game. Also, it is important to note that feedback content types may be combined and are not mutually exclusive. For example, feedback in a serious game may provide both outcome feedback and process feedback by informing the learner that his or her response is incorrect and explaining the steps needed to arrive at the correct answer. Although this may not represent an exhaustive list of

Table 7.1 Feedback content table

Feedback type	Definition	Example
Outcome	Knowledge of Result/Response (Verification, Corrective)	“Your answer was incorrect”
	Knowledge of Correct Result/Response (Correct Response)	“The correct answer was B”
	Answer until Correct (Try Again)	The learner must infer that his/her response is incorrect until a correct response is provided and the learner is allowed to continue
	Percent Accuracy	“You were correct on 75 % of the questions”
	Error Flagging (Location of Mistakes)	“The first part of your answer is incorrect”
	Environmental	When building an electric circuit, the bulb will not light up if the path is broken
	Velocity	“Your performance declined from the previous scenario”
	Normative	“You performed better than 80 % of other students”
	Informational	“Remember to apply Ohm’s law when solving for the required voltage”
	Topic Specific	“If you add a battery in serial, the flow rate increases”
Process (Elaborative, Explanatory) ^a	Response Specific	““A path that is not open” is not entirely correct. A bulb and a battery must be contained in the same closed path in order for a light bulb to light”
	Hints/Cues/Prompts (Worked-out example)	“Here is a hint: A bulb is contained in what?”

Note: Table adapted from Mason and Bruning (2003), Shute (2008), and Van Buskirk (2011). These feedback types are not mutually exclusive. Feedback content may be combined. Types of feedback may also be labeled differently in feedback studies

^aProcess feedback may be conceptual or procedural

feedback content possibilities, it represents our approach to interpreting the variants of feedback used in the experiments discussed throughout this chapter. Below we describe the extant literature on feedback as it relates to serious games using the terminology of the original studies (and the new designation as represented in Table 7.1 where appropriate).

7.3.1.1 Comparing Explanatory Feedback to Corrective Feedback in Science Learning

Explanatory feedback (i.e., process feedback) occurs when a learner performs some action in a game or simulation and receives principle-based feedback describing the quality of his or her performance. Corrective feedback (i.e., outcome feedback), on the other hand, merely informs the learner whether the action he or she took was correct or not. Previous research has demonstrated that explanatory feedback is a very effective strategy to promote learning (Bangert-Drowns et al., 1991; Cameron & Dwyer, 2005; Mayer, 2014a; Shute, 2008).

According to the *feedback principle* in multimedia learning, novices learn more deeply with explanatory feedback than with corrective feedback (Johnson & Priest, 2014). In an exemplary study by Mayer and Johnson (2010), novice college students learned about electric circuits by playing the Circuit Game. The Circuit Game is an arcade-style game that consists of nine training levels, in which students answer questions about the rate of current of circuits in various configurations. A tenth level serves as an embedded near transfer test and students must determine, given two circuit diagrams, which light bulb burns the brightest. One group of students received only corrective feedback about their answers (i.e., corrective feedback group) with either a “ding” to denote a correct answer or a “buzz” to denote an incorrect answer (i.e., knowledge of correct response feedback group). A second group of students received corrective feedback in addition to explanatory feedback (i.e., explanatory feedback group) that showed the correct answer with an explanation of why that answer was correct (e.g., If you add a battery in serial, the flow rate increases; topic specific feedback). Figure 7.2 shows example screenshots from each condition. The authors found that those who received explanatory feedback during training performed better on the nine training levels and on the embedded transfer test than those who received only corrective feedback. They concluded that providing explanatory feedback guides the students to develop a deeper understanding of the material than merely providing corrective feedback.

Another example of the benefits of explanatory over corrective feedback is a set of studies by Moreno (2004). In two experiments, college students learned about botany by playing the Design-A-Plant simulation game. In this game, learners travel to a new planet with certain environmental conditions, and the learner is tasked to design a plant that would survive such conditions. Along the way, the learners are guided by an animated pedagogical agent, who provides feedback to learners on

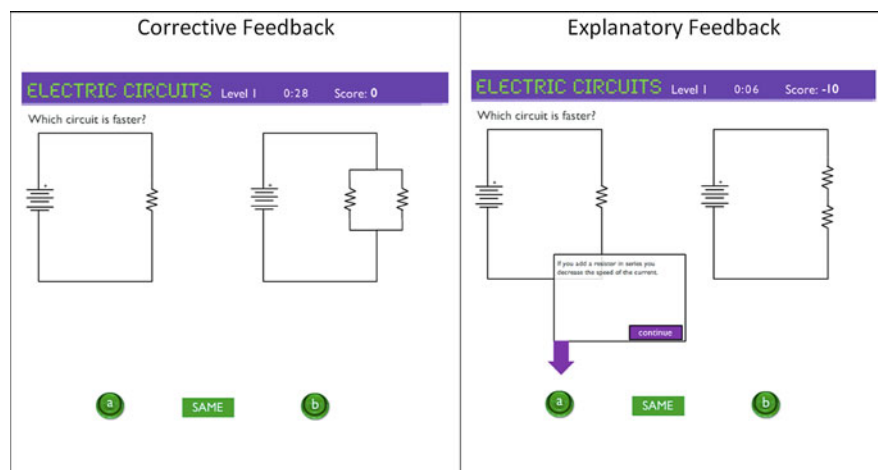


Fig. 7.2 Screenshots from the Circuit Game illustrating the corrective feedback condition (*left*) and the explanatory feedback condition (*right*). Adapted from Mayer and Johnson (2010)

their choices of roots, stems, and leaves that are most appropriate for the eight given environments learners travel to in the course of the game. Following the game, students completed a retention test, in which they had to recall the types of roots, stems, and leaves they learned about; and a transfer test, in which they designed plants for other nonstudied environmental conditions and had to determine, given a plant, the environment for which it was best suited. Students participated in one of two different feedback conditions. In the corrective feedback condition, the pedagogical agent informs the student whether his or her answer was correct or incorrect and then shows the student the correct answer (i.e., knowledge of correct response feedback). In the explanatory feedback condition, students received explanatory feedback from the pedagogical agent that explained why the student's choice of plant part was correct or incorrect (i.e., topic specific feedback). It should be noted that in both cases, the learners receive feedback about the correctness of their answer, but the explanatory feedback condition receives additional information about why the answer was correct or incorrect. Across both experiments, the explanatory feedback condition performed better than the corrective feedback condition on the transfer test, produced fewer incorrect responses during learning, and rated the game as more helpful. These results were replicated in a follow-up experiment conducted by Moreno and Mayer (2005, Experiment 1), and they also found that those who received explanatory feedback, as compared to those who received only corrective feedback, produced significantly more correct explanations for their answers on the transfer test and showed evidence for a reduction in the number of misconceptions. In summary, research evidence demonstrates that providing explanatory feedback that explains why a particular answer is correct promotes learning as compared to only providing corrective feedback.

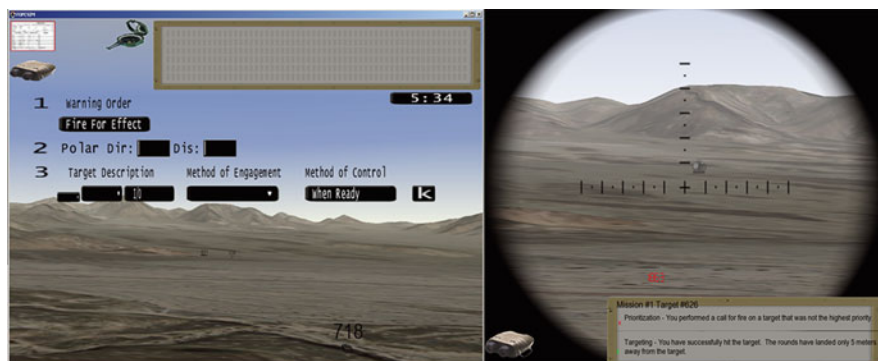


Fig. 7.3 Screenshots from FOPCSIM illustrating the call-for-fire sheet (*left*) used in Astwood et al. (2008), Landsberg, Van Buskirk, and Astwood (2010), and Van Buskirk (2011) and feedback (*right*; showing immediate, visual, outcome feedback for correct targeting and incorrect prioritization) used in Van Buskirk (2011)

7.3.1.2 Comparing Process and Different Types of Outcome Feedback in a Military Decision-Making Simulation

In a study by Astwood, Van Buskirk, Cornejo, and Dalton (2008), college students played a military-based simulation called the Forward Observer PC-based Simulation (FOPCSIM) to train a call for fire task (see left panel of Fig. 7.3). The goal of this task was to destroy enemy targets before they get too close to the learner's position, paying attention to the priority of the targets. The task required three steps: the learner must locate the target, identify the target, and call the position and target type to the artillery or mortar unit. Students were assigned to one of four feedback conditions. In the percent accuracy feedback condition, students were told their overall score, such as "You correctly chose the high priority target 60 % of the time" (i.e., a type of outcome feedback). In the process feedback condition, students received information about how to perform the task, such as "Be sure to use the binoculars to scan the environment and refer to the prioritization rules" (i.e., informational feedback). In the normative feedback condition (i.e., a type of outcome feedback), students received information comparing their performance to that of others, but it was not specific, such as "You were in the 90th percentile. You performed better than 90 % of participants and 10 % performed better than you." Finally, in the control condition, students did not receive any feedback about how they performed the task. The authors found that the process feedback condition outperformed all of the other conditions on prioritization decisions (e.g., determining which targets were the highest priority for targeting actions) and that the normative and percent accuracy feedback conditions did not differ from the no-feedback control condition. They concluded that providing process feedback is an essential step for learning in simulation-based training environments because it focuses trainees on learning decision-making skills (e.g., steps to completing the task) versus performance outcomes (i.e., percent of correct responses).

7.3.1.3 Comparing Different Types of Process Feedback in a Military Procedural Learning Simulation

Billings (2012) and Serge, Priest, Durlach, and Johnson (2013) performed experiments to investigate the impact of different levels of feedback specificity on learning search and report procedures in a game-based simulation. In these experiments, college students learned about procedures for searching rooms for objects and personnel and the protocol for making reports to headquarters of what they found. Students then practiced these procedures in the game across three training scenarios and one performance scenario that was more complex with time pressure and low lighting conditions. The level of feedback specificity was manipulated between groups. In the general feedback condition, students received high-level feedback at the terminal learning objective level about their performance at the end of each training scenario (i.e., informational feedback), such as “Remember to apply the procedures for entering and exiting buildings.” In the detailed feedback condition, students received more specific feedback at the enabling learning objective level about their performance at the end of each training scenario (i.e., topic specific feedback), such as, “Before entering or tagging a building, you should walk around the entire building to make sure it is not already tagged.” In the outcome feedback condition, students received their performance score at the end of each training scenario (i.e., percent accuracy feedback). In the performance scenario, students in the detailed feedback condition outperformed those in the outcome feedback condition. In the Billings (2012) experiment, the general feedback condition performed similarly to the outcome feedback condition, and neither group showed any performance improvement across the four scenarios. To follow up on this finding, Serge et al. (2013) allowed students in the general feedback condition the option to review the training manual after receiving feedback but before starting a new training scenario (i.e., there were three opportunities to review the training manual) to remind themselves of the specific search and report procedures. The results showed that students who opted to review the manual performed at the same level as the detailed feedback condition, whereas those who did not review the manual performed as poorly as the outcome feedback condition. These results support the notion that students in the general feedback condition may have forgotten some of the procedures, and the high-level information provided in the feedback was not enough to help them repair their errors in subsequent scenarios. Therefore, both Billings (2012) and Serge et al. (2013) concluded that providing detailed feedback during training improves performance.

Taken together, the results of these experiments demonstrate that more detailed process feedback improves performance and learning outcomes for novice learners when compared to outcome feedback, consistent with the feedback principle in multimedia learning. Providing more specific feedback helps to reduce a learner’s extraneous cognitive processing, by guiding them to the correct action or answer within the game, leading to higher learning outcomes and performance. Additionally, generative processing may be affected by detailed feedback in that this form of processing requires motivation to make sense of the material. Essential processing may not be affected by providing detailed feedback as feedback should not inherently change the difficulty of the material to be learned.

7.3.2 *Modality of Feedback*

The modality in which feedback is presented can have a large impact on the effectiveness of the feedback. The modality principle states that people learn more deeply when words are presented in spoken form rather than printed text in a primarily visual task (i.e., when the limited capacity of the visual channel is already occupied by visual information; Mayer, 2009, 2014a, 2014b). The theoretical rationale for the modality principle is that when learners are playing a visually based game and information is presented as printed text, their visual channel can become overloaded. By presenting that information in spoken form, the learner can offload some of the demands on the visual channel by utilizing the verbal channel to process the words. This allows learners to increase their cognitive processing capacity by utilizing both channels to process the incoming essential information; this is an example of managing essential processing.

The modality principle is well established in the multimedia learning literature (Ginns, 2005; Low & Sweller, 2014; Mayer & Moreno, 1998; Mayer & Pilegard, 2014; Mousavi, Low, & Sweller, 1995), and research evidence is beginning to show that it holds true for serious games. For example, Moreno, Mayer, Spire, and Lester (2001) performed a series of experiments examining the attributes of pedagogical agents that lead to deeper learning using the Design-A-Plant simulation game that was discussed earlier (Sect. 7.3.1.1). The authors examined how the modality of feedback and instructions delivered by an animated pedagogical agent (Experiment 4) or a human agent presented in the form of a video (Experiment 5) affects learning. One group received feedback and instructions from the agent in spoken form, while the other group received that information as onscreen text. Across both experiments, the authors found evidence for a modality effect, such that the spoken text group outperformed the onscreen text group on tests of retention and transfer and rated the material as more interesting. They concluded that students could devote more cognitive resources to meaningful learning by utilizing both the verbal and visual channels, thereby managing essential processing.

Using a more complex task, Fiorella, Vogel-Walcutt, and Schatz (2012) tested the modality principle in an interactive military simulation training a decision-making task. College students performed in two call-for-fire scenarios (a visual-spatial task), in which they had to decide the appropriate warning orders to give when firing on an enemy, taking into account the expense of ammunition chosen and level of damage to targets. The correct choice would be a warning order that costs less and is less destructive for vehicle targets or is more destructive and expensive for tank targets. Two groups of participants received real-time feedback on their performance either as onscreen text or spoken form, while a third group received no performance feedback, serving as a control. The authors found that the spoken feedback group performed better than the text feedback and control groups during the more complex training scenario and on an assessment scenario in which no feedback was provided. On retention tests of procedural knowledge, those in the spoken feedback group scored higher than the text feedback group, although there

were no significant differences in scores for tests of declarative or integrative knowledge. The authors concluded that, consistent with CTML, when completing a visual-spatial task, real-time feedback should be spoken rather than presented as text to allow the learner to utilize both the verbal and visual channels to process the information. This frees up cognitive resources to engage in more productive cognitive processing to achieve meaningful learning. At the same time, providing real-time feedback as onscreen text overloads the learner's visual channel, leaving the learner with fewer cognitive resources to process the information, and he is unable to engage in the essential and generative processing necessary to achieve meaningful learning.

Another consideration to take into account when designing spoken feedback messages in serious games or simulations is the type of voice used: synthetic or natural speech. Synthetic speech is often reported in experiments as qualitatively different than natural speech (Paris, Thomas, Gilson, & Kincaid, 2000), leading individuals to encode the information differently. Likewise, Smither (1993) suggested that synthetic speech places more processing demands on a person than natural speech, which according to CTML would elicit extraneous cognitive processing on behalf of the learner and may lead to a learning decrement. In fact, Mayer and DaPra (2012) found that college students performed best on a transfer test when the animated pedagogical agent had a human voice versus a machine voice. They also found that other human-like features of the agent, such as eye gaze, facial expression, and body movements facilitated deeper learning, consistent with social agency theory.

In summary, the research on modality supports the idea that for a visual task, feedback is more effective when presented as spoken words (i.e., human speech). When playing a serious game with a highly visual component, presenting feedback as spoken words allows learners to utilize both their verbal and visual channels for processing the incoming information, thereby increasing their cognitive processing capabilities. Learners then have more cognitive resources available to devote to essential and generative processing, leading to better learning outcomes. However, this guidance does not suggest that feedback always be presented as spoken words, but merely to consider the nature of the game and the limits on a learner's cognitive system and to present feedback in the modality that is most appropriate.

7.3.3 Timing of Feedback

The timing of feedback presentation (i.e., immediate versus delayed) can also have an impact on learning and performance in serious games and simulations. However, guidance on when to give feedback is mixed due to the conflicting theories and research on the timing of feedback (Shute, 2008; Van Buskirk, 2011). For instance, those who argue for the use of immediate feedback suggest that it prevents errors from being encoded into memory (Bangert-Drowns et al., 1991). Temporal contiguity research suggests that when feedback is presented to a trainee immediately, the correct cue-strategy associations in the learning environment are strengthened and

incorrect ones are weakened, thus resulting in better performance (Anderson, Corbett, Koedinger, & Pelletier, 1995; Corbett, Koedinger, & Anderson, 1997).

Alternative theories for feedback presentation suggest that when feedback is presented immediately after an error, the incorrect response interferes with learning the correct answer and learning the correct way to do the task (known as the perseveration-interference hypothesis). Conversely, errors made early on in learning may be forgotten if feedback is delayed, and the removal of “interference” increases the chances that learners will learn the correct information (Kulhavy & Anderson, 1972). Further, proponents of delayed feedback presentation also argue that immediate feedback serves as a crutch. In other words, learners come to rely on the feedback and when it is removed, their resulting performance suffers.

Kulik and Kulik (1988) performed a meta-analysis to compare the effectiveness of immediate and delayed feedback on verbal learning tasks in the laboratory and in the classroom. Their results showed that for applied studies, the use of immediate feedback yielded performance improvements over delayed feedback. However, results from laboratory studies showed the opposite effect, indicating an advantage of delayed feedback. In a recent study, Johnson, Priest, Glerum, and Serge (2013) investigated the timing of (detailed) feedback delivery in a game-based simulation for training search and report procedures that was discussed earlier (see Sect. 7.3.1.3). Participants received feedback according to three feedback timing schedule conditions: immediate (after an error), chunked (at a logical stopping point in the scenario), or delayed (at the end of a scenario). Although the authors found no statistically significant differences between the timing conditions, the trends in the data showed that the immediate feedback groups scored marginally higher than both the delayed and chunked groups. Further, they found that delayed feedback groups reported increases in cognitive load over time, while the chunked and immediate groups reported decreases in cognitive load. The authors concluded that providing immediate feedback may serve to reduce extraneous cognitive load for novices when learning procedures in a serious game. The results from this experiment are not overwhelming, yet they suggest some preliminary evidence for the use of immediate feedback in serious games. Certainly more research is needed to determine the optimal feedback timing schedules to enhance learning and performance in serious games and simulations.

In addition to the conflicting theoretical perspectives, the timing of feedback literature is riddled with inconsistencies due to several reasons. First, there is no agreement on the operational definitions of what is considered delayed versus immediate feedback. For example, most researchers would agree that feedback presented after a test item or a trial that lasts between 15 and 30 s is immediate feedback. Likewise, most researchers would agree that feedback presented days after performance would be considered delayed. However, researchers disagree on operational definitions in the middle ground. For instance, is feedback presented after an entire test (with multiple test items) or after the completion of a game or scenario that lasted 15 min (with several stimulus–response actions) considered immediate feedback or delayed feedback? Second, many of the studies in this domain contain confounds, such as providing a double stimulus exposure (Kulik & Kulik, 1988) and not controlling the

amount of feedback given. For example, some researchers provide less feedback when given immediately, because they do not want to overload the participant during game performance. Last, the feedback content and other parameters, such as task characteristics, vary greatly in the feedback timing studies. Therefore, more systematic research is needed in this area to investigate the incremental effectiveness of the complex relationships between these feedback variables (e.g., content, timing, task parameters, modality, etc.). One experiment that combined feedback variables (Van Buskirk, 2011) investigated the relationship of feedback timing (immediate, delayed), content (outcome, process), and modality (visual, auditory) within the Forward Observer PC-based Simulation (FOPCSIM) described earlier as the Marine Corps call-for-fire task (a highly visual task; see left and right panel of Fig. 7.3). The results showed that participants who received immediate, auditory, process feedback outperformed all other groups on target prioritization performance. Studies taking a value-added approach to research such as this will help provide guidance for practitioners about what combinations of feedback parameters are effective for learning.

7.3.4 Individual Differences

Consideration for individual differences of the learner can lead to optimized feedback strategies when developing a serious game or simulation. Characteristics of individuals can affect the efficacy of feedback on learning outcomes and may interact with other individual differences or the type of feedback given (Narciss et al., 2014). Narciss et al. (2014) summarize factors that are typically addressed in feedback strategy research, including (a) prior knowledge, (b) motivation, (c) gender, and (d) meta-cognitive skills (e.g., self-assessment, help-seeking). Yet another way to characterize individual difference variables is to consider whether they are developed or stable learner characteristics (Vandewaeter, Desmet, & Clarebout, 2011). Developed learner characteristics, such as domain knowledge and task motivation, could change as a result of the learning process. More stable learner characteristics include gender, intelligence, spatial ability, and some have argued learning style (but see Pashler, McDaniel, Rohrer, & Bjork, 2008). Although there is a developing body of literature on individual differences in feedback, the research on how learner characteristics interact with feedback in serious games or training simulations remains sparse (Narciss et al., 2014; Timmers, Braber-van den Broek, & van den Berg, 2013).

Considering domain knowledge, a developed learner characteristic, evidence from multimedia learning research has shown that experts and novices differ in the amount of instructional support necessary to foster meaningful learning. The *expertise reversal effect* is the finding that instructional support that is helpful for novice learners may actually hinder learning of experts in that domain (Kalyuga, 2014; Kalyuga, Ayres, Chandler, & Sweller, 2003). There have been few studies examining this result in the feedback and gaming literatures in particular. One study by Smits, Boon, Sluijsmans, and van Gog (2008) investigated feedback content, timing, and

learner characteristics (prior knowledge) on learning outcomes in a web-based environment teaching genetics. They varied feedback content by offering either global (i.e., outcome and informational feedback) or elaborated (i.e., outcome, informational, the worked-out solution, and topic specific feedback) information about students' answers. Feedback timing was also varied: presented either immediately following each task or delayed to every other task. The authors found no effect for feedback timing regardless of prior knowledge level. For learners with low prior knowledge, there was no effect of feedback content on learning outcomes; however, for learners with higher prior knowledge, those who received global feedback outperformed those who received elaborated feedback. This finding suggests that learners with higher prior knowledge of a task may need less detailed feedback.

Considering spatial ability, a generally more stable learner characteristic, Moreno and Mayer (1999) found that the effectiveness of process feedback was influenced by the learner's spatial ability. In their experiment, elementary school children learned addition and subtraction playing an interactive multimedia program. For students who received process feedback, those with higher spatial ability had better learning gains from pre- to posttest than those with lower spatial ability. The authors reasoned that this difference in the effectiveness of process feedback is due to learners with lower spatial ability being cognitively overloaded during learning and not having the cognitive resources necessary to process the information in the feedback. On the other hand, the students with high spatial ability did have the cognitive resources available to process the feedback, and as a result they demonstrated higher learning gains. This finding supports the notion that spatial ability may be an important learner characteristic to consider when developing interactive learning environments for teaching complex skills, but certainly more research is needed to fully explore the relationships between spatial ability and type of feedback in various contexts.

Considering gender, another stable learner characteristic, Landsberg et al. (2010) investigated the relative efficacy of process (i.e., topic specific), velocity, and no feedback in FOPCSIM, the call for fire simulation (see left panel of Fig. 7.3). In this study, the results showed that gender significantly interacted with feedback condition on target prioritization performance, meaning that the most useful feedback was different for males and females. Specifically, they found that males performed better (i.e., correctly selected the highest priority target) when they received velocity feedback compared to process feedback, although males in the no feedback control group were not significantly different from either process or velocity feedback groups. Conversely, females performed significantly better when they received process feedback as compared to no feedback. These results suggest that process feedback helped females more accurately prioritize, while velocity feedback helped males more accurately prioritize. Furthermore, they found that females who received process feedback significantly outperformed males who received process feedback on target prioritization. This research provides evidence for the notion that a stable learner characteristic, such as gender, may influence the effectiveness of process feedback in serious games and simulations.

Finally, Narciss et al. (2014) sought to determine the effect and interaction of both stable and developed learner characteristics (e.g., gender, intrinsic motivation)

on the efficacy of feedback on problem-solving behavior of sixth and seventh grade boys and girls in a computer adaptive math-learning program. The results showed mixed effects and interactions of learner characteristics with feedback content on learning and behavior outcomes. Overall, they found that there were differences between boys and girls in how much feedback impacted learning outcomes, such that boys benefited less from feedback on practice tasks, had lower knowledge gains from pre- to posttest, and their perceived competence declined. In contrast, girls did have significant gains on the knowledge posttest.

Taken together, these studies provide evidence that serious games designers and researchers should consider relevant individual differences for specific educational and training domains and how those learner characteristics may interact with different feedback strategies. More specifically, stable learner characteristics, such as gender and spatial ability, may have an impact on the efficacy of process feedback in serious games. Additionally, there is some evidence that developed learner characteristics may also impact the effectiveness of process feedback, such as level of domain expertise, but more systematic research is needed to understand which variables may play a role in which domains.

7.4 Discussion

7.4.1 *Summary of Findings*

In this chapter, we assessed the feedback literature in serious games from a value-added perspective, identifying key findings to arrive at the most applicable feedback characteristics benefiting learning outcomes. The majority of the studies address the effects of feedback content, modality, timing, and/or adaptation to individual differences as possible strategies for better learning through serious games. For each of these feedback features, we summarize the findings below:

1. *Content*

The content of feedback refers to the type of information presented to learners. Feedback content may be combined in numerous ways depending on the learning objectives and learner characteristics. We grouped these feedback types into outcome-based feedback (informing learners about their performance outcome or learning progress) or process-based feedback (directing learners on the process to reach the correct answer or progress). Overall, research on feedback content seems to suggest that process (explanatory) feedback is more beneficial than outcome (corrective) feedback, because providing a deeper explanation that guides learners to the correct answer and reasoning behind it rather than merely informing learners about the correctness of their performance reduces extraneous processing and may enhance essential processing, leading to higher learning outcomes. However, there is some evidence that individual differences may impact efficacy of process feedback (see below).

2. *Modality*

Feedback modality is the way in which the feedback information is presented. For example, feedback could be visual or auditory/spoken, delivered by a pedagogical agent or video of a person, and many other forms of presentation. Although conclusions at this point are tenuous as these studies vary widely in terms of educational domain and type of game or simulation, determining the best modality for feedback presentation appears to depend largely on the design of the instructional material. Good design should take into consideration the amount of cognitive load placed on the learner such that the modality of feedback does not compete with the instructional material for limited mental resources. That is, for a highly visual-spatial game, presenting feedback as spoken text allows learners to utilize both the visual and verbal channels to process the information.

3. *Timing*

The literature on timing of feedback in serious games is sparse, but a trend may be developing to indicate an advantage for immediate feedback for acquisition of learning material, although how feedback timing affects long-term retention is still unclear. Further, feedback timing research indicates that this variable may have complex relationships with other feedback features (i.e., content and modality) and should not be investigated in isolation.

4. *Adaptation to Individual Differences*

In general, the interaction between feedback strategies and characteristics of the individual learner is not well known. However, several research studies suggest that stable learner characteristics (e.g., gender, spatial ability) may impact the effectiveness of different feedback types. Although results on the relationship between developed learner characteristics and feedback variables are scant, initial findings suggest that level of prior knowledge can impact the effectiveness of process feedback.

7.4.2 *Future Research Directions and Conclusions*

This review highlights the lack of systematic research on feedback in serious games necessary to define the parameters of feedback strategies that optimize learning outcomes. The studies described throughout this chapter demonstrate how the literature on feedback in simulations and games varies greatly on terminology used, characteristics of feedback given, educational or training domain, types of learning outcomes, and the game itself. Further, the differing feedback features are often not described in enough detail to support replication and a comparison of results between studies (Hatala, Cook, Zendejas, Hamstra, & Brydges, 2014; Ke, 2009; Merchant et al., 2014). Therefore, authors should be mindful to provide examples and explicit details of the feedback and task they use in their research so that future researchers can make comparisons and draw conclusions from the large bodies of research in serious games and simulations. Additionally, in order to develop

evidenced-based principles and guidelines for practitioners, we call for a value-added approach to future research in this domain to understand the relationships between feedback variables (content, modality, and timing) and learner characteristics (developed and stable). Lastly, many of the research findings presented were based on performance improvements during acquisition (with the exception of Moreno, 2004; Mayer & Moreno, 2005). However, future research should focus on long-term retention and transfer. This is not a new concept (see Schmidt & Bjork, 1992), but one that has ramifications for research in this area. The goal of creating educational and training materials is to produce lasting effects on knowledge, skills, or behavior, and the focus of current feedback literature on acquisition as opposed to retention limits the conclusions we can draw from studies of feedback parameters. In fact, a study by Astwood and Smith-Jentsch (2010) found that while immediate feedback groups had higher performance improvements during acquisition, the complete opposite results were found showing an advantage for delayed feedback during a retention phase.

Accordingly, more research is needed to determine the effects of feedback on learners' motivation when playing serious games. Although many researchers extol the virtues of games for promoting motivation, there is little empirical evidence to validate these claims. One recent meta-analysis even concluded that playing serious games (in general) may not be more motivating to students than traditional teaching methods (Wouters et al., 2013). We recommend that future researchers take a value-added approach to studying the effects of feedback (and other instructional strategies) on motivation and learning in serious games. Such research is necessary to develop an understanding of the mechanism by which motivation affects cognitive processing, which instructional strategies are most effective at fostering motivation, and whether particular individual difference variables enhance (or diminish) motivation.

Given that a learner's cognitive system has limited capacity, designers should be mindful of how feedback variables impose processing demands. From a review of the literature, we were able to identify features of feedback that could reduce cognitive processing demands to achieve improved learning and performance in serious games and simulations: feedback content, modality, timing, and adaptation to individual differences. However, there is still a wealth of systematic research that is needed in order to provide evidenced-based design principles for serious games and simulation designers.

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